

Replication Paper Quantitative Methods II

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1. Introduction

In the following analysis, I will replicate "Are popular and powerful committees more representative? Evidence from the ninth European Parliament" by Steffen Hurka and Constantin Kaplaner. In their paper, the authors assess the representativeness of the committees of the 9th European Parliament (EP) (2019-2024). More specifically, they assess the representativeness of entire committees by comparing their preference distributions with those they find in the plenary on the left/right and pro/anti EU dimension. For their analysis, the authors combine new data on committee memberships in the ninth EP with data on policy preferences of national parties from the Chapel Hill Expert Survey (CHES). CHES estimates party positioning on ideology and policy issues, and international relations for national parties in countries across the world. In order to assess committee representativeness, the authors thus calculate the extent to which the preference distributions of committees and the plenary overlap, showing that committee representativeness is a function of committee popularity and power.

Assessing the representativeness of committees in the EP is both timely and relevant. The EP is the second largest elected legislative body in the world, with legislative outcomes affecting over 450 million citizens. Committees play a key role in the European legislative process, as committee positions strongly predetermine the policy position the EP will later adopt in plenary.

I chose this paper since my PhD research topic focuses on measuring interest group influence in the European Parliament at both the committee and plenary stage. Committee composition and representativeness thus represents a key element in my upcoming research, as by party and nationality of the Committee members may strongly affect how much access is given to interest groups.

2. Original Study: Hurka and Kaplaner's Analysis

The author's analysis relies on a dataset that combines full committee memberships in the 9th EP with data on policy positions of the Members of Parliament's (MEP) national political parties from the CHES. Based on theoretical foundations, the authors test the following hypothesis: *More popular and more powerful committees are more representative of the plenary on key dimensions of political conflict than less popular and less powerful committees.*

The authors test the hypothesis for the general left/right dimension, ranging from 0 (extreme left) to 10 (extreme right) and the pro/anti EU dimension, ranging from 1 (strongly opposed to European integration) to 7 (strongly in favour).

Dependent variable: the authors measure committee representativeness by calculating the overlap between the committees' density distributions and the plenary's density distribution using the overlapping function in R. The authors also clarify that in the scope of the article, they only focus on the analysis of preference overlap. To do so, they create a dataset containing the full committee memberships of each MEP in the 9th EP. Each MEP was assigned the position of their respective national party on the left/right and pro/anti EU dimension respectively based on CHES data.

To illustrate the benefits of this approach, the authors randomly generated two distributions with the same mean, median and standard deviation (Figure 1). The first distribution was forced into a bimodal shape, while the second distribution is normally distributed. Although one can observe vastly different shapes, the

distributions would be considered perfectly representative of each other if one only relied on measures of central tendency. Rather, the overlap of the distributions better captures the relevant differences.

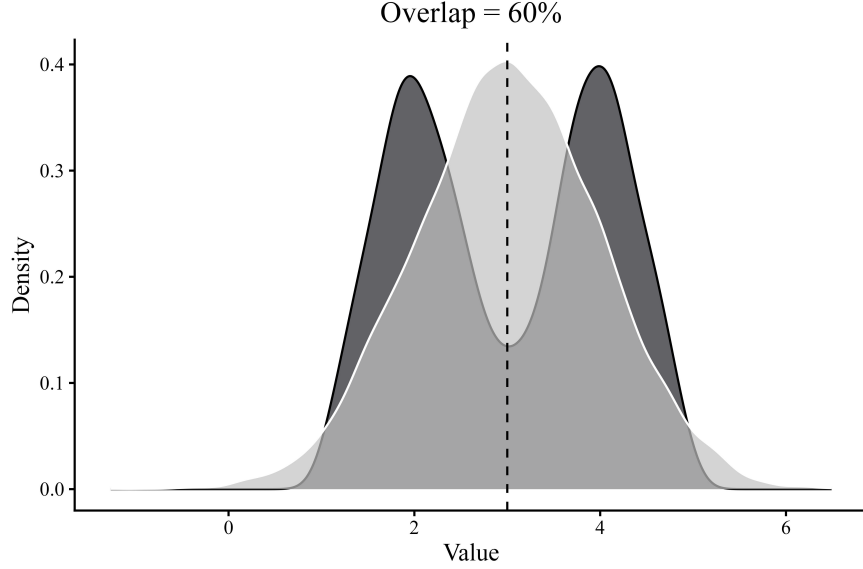


Figure 1: Simulation of two distributions with the same mean, median and standard deviation

Independent variables: The authors utilise the power and popularity variables as independent variables.

- **Committee power:** The authors adopted an existing categorization by Yordanova (2009), distinguishing committees as more or less powerful based on their respective influence on co-decision legislation and the EU budget.
- **Committee popularity:** Here the authors relied on the survey of Whitaker (2019), in which he asked MEPs to name their most preferred committee. They then use both the percentage of mentions per committee and the resulting ordinal ranking of committees as indicators of committee popularity.

Methods: In their empirical analysis, Hurka and Kaplaner rely on two complementary strategies. First, they inspect deviations from representativeness in a descriptive manner. Second, they estimate the relationships between committee popularity, power and representativeness with ordinary least squares (OLS) regressions. Furthermore, they assess the influence of individual committees on the estimated coefficients, demonstrating that the main findings are not heavily influenced by outliers. In the analysis, the authors only consider full members of the committee, since substitute members may only take part in committee votes when the full member is absent.

3. Replication

3.1. Assessing the data

The main dataset (ep9) is comprised of 1778 observations of 20 variables. The following variables are of interest for the subsequent analysis:

- "committee" and "committee_short": Full and abbreviated names of committees. Nominal.
- "type": Type of committee membership of the MEP (substitute or member). Nominal.
- "id": ID of the MEP assigned by the European Parliament. Nominal.
- "lrgen": The position of the MEP's national party on the general left/right scale, derived from its most recent measurement. Scale of 0 (= extreme left) to 10 (= extreme right).

- "position": The position of the MEP's national party on the pro-/anti-EU scale. Scale of 1 (= strongly opposed) to 7 (= strongly in favour).

3.2. Analysis Replication

First, the authors compare the distribution of preferences on the pro/anti EU dimension in two committees (Foreign Affairs (AFET) and Fisheries (PECH)) with plenary-level distributions. To achieve this, they create new datasets for the analysis: plenary and committee, reduced to the relevant variables (see above). They then proceed to plot the distribution of pro/anti-EU stance of the MEPs per committee (light grey) against the distribution of MEPs in the plenary (dark grey). The following figure illustrates the distribution of pro/anti-EU MEPs for the Foreign Affairs and Fisheries Committees:

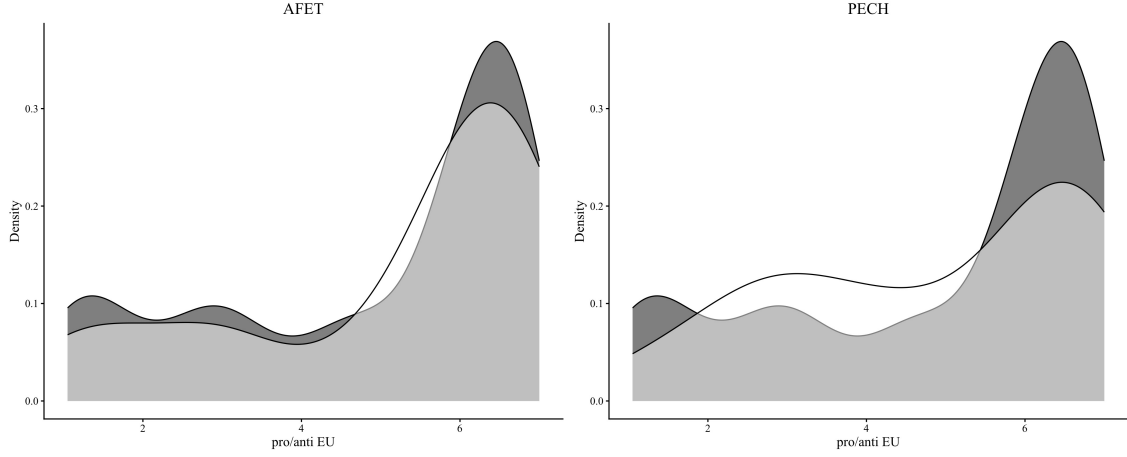


Figure 2: Representativeness of the EP's Foreign Affairs (AFET) and Fisheries (PECH) committees on the pro/anti EU dimension

The authors then proceed to analyse the level representativeness across all the EP's committees. To do so, they plot the committees on left/right, pro/anti-EU and either more/less powerful or more/less popular with MEPs. High scores/overlap with the plenary-level distribution indicates representativeness of the committees.

Figure 3(a) below thus illustrates how representativeness varies across EP committees and illustrates the extent to which more and less powerful committees differ from each other regarding left/right and pro/anti-EU positioning. The scatterplot reveals that the 10 committees classified as "non-powerful" occupy about the same range as the more powerful committees on the left/right dimension. On the pro/anti EU dimension, in contrast, we find that powerful committees tend to be more representative than less powerful committees, with the notable exceptions of Legal Affairs (JURI, powerful) and Foreign Affairs (AFET, not powerful).

Figure 3(b) plots representativeness of each committee against perceived popularity by MEPs. Here, increased representativeness of more popular committees can be observed. Less popular committees seem to dominate the lower overlap (Legal Affairs (JURI), Petitions (PETI), and Budgetary Control (CONT)), indicating lower representativeness.

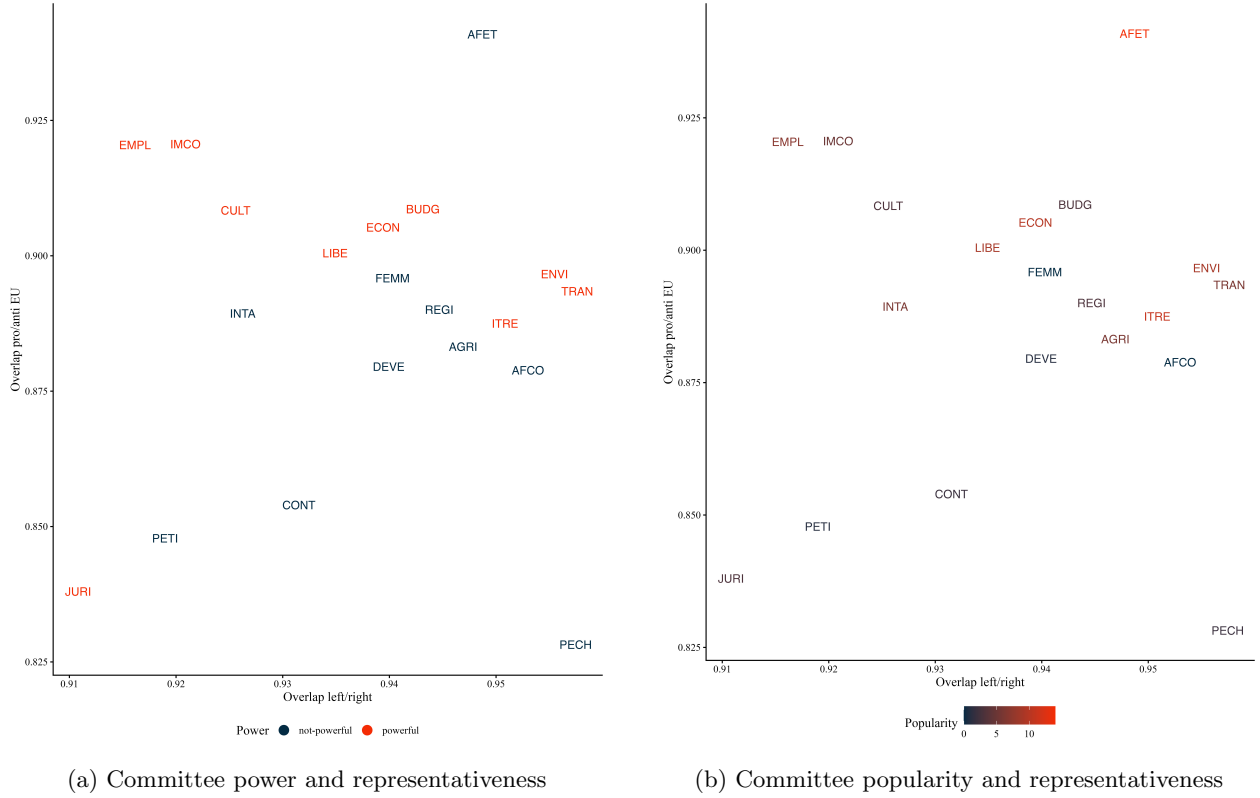


Figure 3: Overall committee representativeness by power and popularity of the committee

The authors then proceed with the following regressions. Since the analysis only has 20 cases, the authors only estimate one parameter per regression and also test the extent to which the coefficients are sensitive to the exclusion of individual committees.

- **Model I (left/right):** Regresses committee representativeness on the left–right dimension on committee popularity (percentage measure).
- **Model II (left/right):** Regresses committee representativeness on the left–right dimension on committee popularity using an ordinal rank measure.
- **Model III (left/right):** Regresses committee representativeness on the left–right dimension on committee power.
- **Model IV (pro/anti EU):** Regresses committee representativeness on the pro/anti EU dimension on committee popularity (percentage measure).
- **Model V (pro/anti EU):** Regresses committee representativeness on the pro/anti EU dimension on committee popularity using an ordinal rank measure.
- **Model VI (pro/anti EU):** Regresses committee representativeness on the pro/anti EU dimension on committee power.

Table 1: Ordinary least squares regression

	I (left/right)	II (left/right)	III (left/right)	IV (pro/anti EU)	V (pro/anti EU)	VI (pro/anti EU)
Committee popularity	0.071 (0.084)			0.406** (0.140)		
Committee popularity (rank)		-0.030 (0.059)			-0.270* (0.098)	
Committee power			-0.529 (0.651)			1.917 (1.228)
Constant	93.456*** (0.536)	94.105*** (0.667)	94.077*** (0.460)	86.783*** (0.890)	91.498*** (1.114)	87.877*** (0.868)
R^2	0.037	0.014	0.035	0.318	0.295	0.119
N	20	20	20	20	20	20

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

It has to be highlighted that the coefficients above show a slight divergence with the coefficients of Hurka and Kaplaner's coefficients. While the overall trends remain the same, for instance the estimated effect of committee popularity on representativeness in Model IV is smaller in the replication (0.406) than in the original study (0.540). Additionally, the Model V statistical significance drops from a $p < 0.05$ level to a $p < 0.1$ level. These discrepancies are likely due to differences in the construction of the overlap measure, as I had to make manual adjustments to the R code. Additionally, the original study assigns party group means to missing observations, whereas my replication relies on listwise deletion. Given the small sample size ($N = 20$), even minor differences in data processing can lead to noticeable changes in coefficient estimates. Importantly, however, the direction of the main effects are consistent with the original findings.

In accordance with the descriptive evidence provided in the paper, the authors find that neither committee popularity nor power helps explain deviations in committee representativeness on the left/right dimension (Models I–III). However, committee popularity increases committee representativeness significantly on the pro/anti-EU dimension (Model IV). Every percentage point increase in popularity is associated with an increase in overlap on the pro/anti-EU dimension of .4 percentage points. The effect also holds if we use an ordinal ranking of committee popularity resulting from Whitaker's survey (Model V, note that higher values indicate lower popularity). Overall, the findings show that one may estimate the overlap on the pro/anti EU dimension to decrease by roughly 6 percentage points between the most popular and the least popular committee. While these effects do not appear large, the intercept indicates that there is a rather high baseline representativeness. Accordingly, one must interpret the coefficient against the background that committees can impossibly become completely unrepresentative in part due to the proportionality principle at the party group level.

Model VI does not show a significant relationship between committee power and representativeness on the pro/anti EU dimension, which may be due to the two outlying cases of AFET and JURI. The authors highlight that upon exclusion, the model turns significant.

Finally, the authors proceed to test the significance of their findings, by employing the logic of a jackknife 'leave-one-out' procedure. The figures below show the results of said approach. Figure 5 shows that the significance of committee popularity is not heavily affected by individual committees. Only the exclusion of AFET impacts on the significance of the association. Additionally, Figure 6 demonstrates that the exclusion of JURI and AFET impacts on the power estimate.

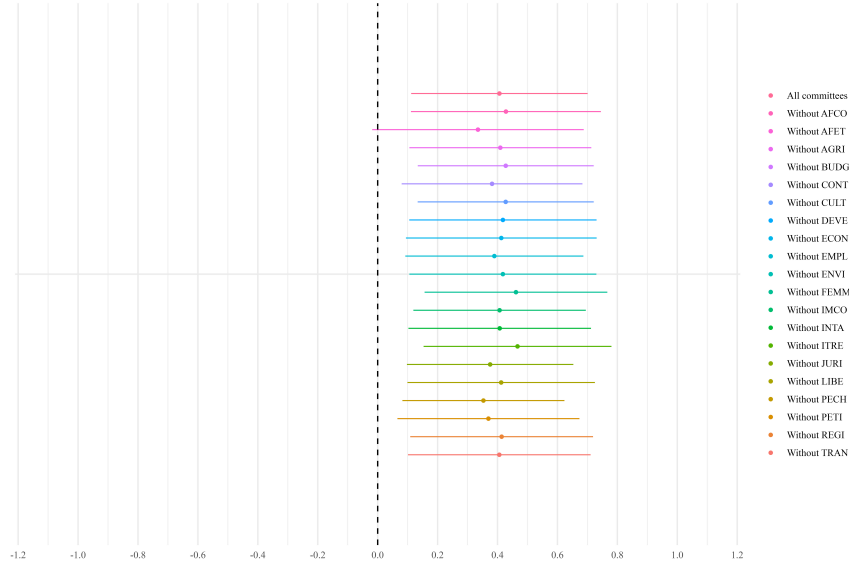


Figure 4: Committee popularity and representativeness (Model IV, pro/anti EU). Note: Regression coefficients estimated in model IV (Table 1) and their 95% confidence intervals.

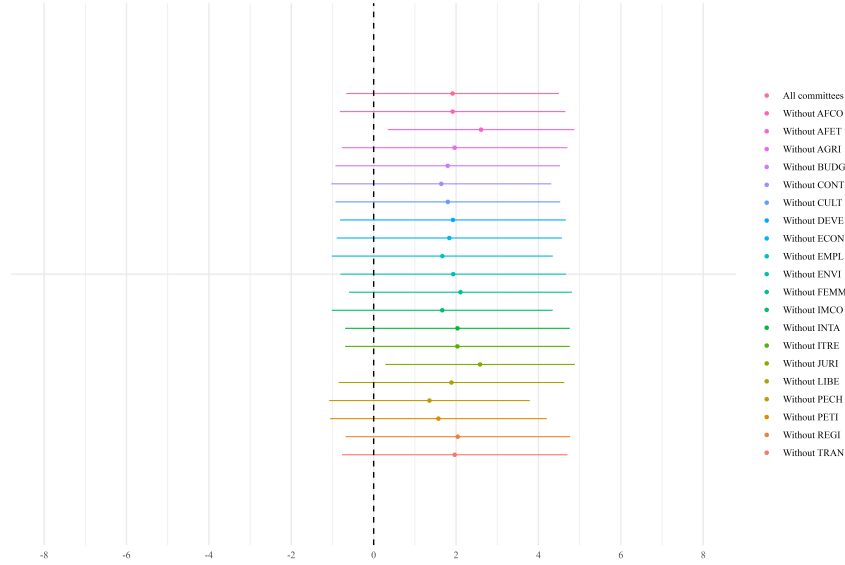


Figure 5: Committee power and representativeness (Model VI, pro/anti EU). Note: Regression coefficients estimated in model VI (Table 1) and their 95% confidence intervals.

3.3. Conclusion of replication

The replication broadly confirms the main findings of Hurka and Kaplaner (2020). In line with the original study, no systematic relationship is found between committee characteristics and representativeness on the left–right dimension. By contrast, committee popularity is positively associated with representativeness on the pro/anti EU dimension, whereas the effect of committee power remains weaker and less consistent. While small differences in coefficient estimates emerge, these do not alter the substantive conclusions. Overall, the results support the authors’ argument that variation in committee representativeness is dimension-specific and primarily driven by committee popularity rather than institutional power.

4. Extension

4.1. Adding an interaction term: committee power and popularity

As an extension, to further explore the relationship between committee popularity and representativeness, I decided to estimate a model including an interaction between popularity rank and committee power on the pro/anti-EU dimension.

```
model_interaction <- lm(overlap_position ~ popularity * powerful, data = reg_base)
summary(model_interaction)
```

The model provided me with the following output:

Table 2: Interaction model: committee popularity and power (pro/anti EU)

	(I) Overlap (pro/anti EU)
Committee popularity	0.494* (0.189)
Committee power (powerful)	2.887 (2.160)
Popularity $\times$ power	-0.379 (0.325)
Constant	86.148*** (1.015)
Observations	20
R^2	0.387
Adjusted R^2	0.272
Residual Std. Error	2.430 (df = 16)
F Statistic	3.367 (df = 3; 16)

Note: Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.1.1. Extension result interpretation

The interaction model using the continuous measure of committee popularity provides further insight into the relationship between popularity, committee power, and representativeness on the pro/anti EU dimension. The results show that committee popularity has a positive and statistically significant effect on representativeness ($\beta = 0.494$, $p < 0.05$), indicating that more popular committees tend to better reflect the distribution of preferences in the plenary. Substantively, a one-unit increase in popularity is associated with an increase of approximately 0.5 points in overlap.

By contrast, the coefficient for committee power is positive but not statistically significant ($\beta = 2.887$, $p = 0.200$), suggesting that powerful committees may appear more representative, but this difference cannot be distinguished from random variation. The interaction term between popularity and committee power is negative and statistically insignificant ($\beta = -0.379$, $p = 0.261$), indicating no strong evidence that the effect of popularity differs between powerful and non-powerful committees.

Overall, the model explains a moderate share of the variation in representativeness ($R^2 = 0.387$, adjusted $R^2 = 0.272$) and is statistically significant at the 5% level ($F(3, 16) = 3.367$, $p = 0.045$). These findings reinforce the main conclusion of the original study that committee popularity is an important determinant of representativeness on the pro/anti EU dimension, while committee power and its interaction with popularity play a more limited role.

4.2. Conclusion

Overall, the replication successfully reproduces the central findings of Hurka and Kaplaner (2020). Consistent with the original study, committee characteristics do not appear to systematically explain representativeness

on the left–right dimension. By contrast, committee popularity is positively associated with representativeness on the pro/anti EU dimension, while the effect of committee power remains weaker and less robust. Although some minor differences in coefficient estimates emerge, these do not alter the substantive conclusions. The replication therefore confirms the authors’ main argument that variation in committee representativeness is dimension-specific and primarily driven by committee popularity rather than institutional power.

Building on this, the extension analysis explored whether the relationship between popularity and representativeness differs depending on committee power by introducing an interaction term. While the interaction effect is not statistically significant, the results suggest a substantively meaningful pattern: the association between lower popularity and lower representativeness appears stronger for powerful committees than for less powerful ones. This is consistent with the theoretical expectation that party groups face stronger incentives to ensure balanced representation in more influential committees. However, given the limited sample size and resulting lack of statistical power, these findings should be interpreted with caution.

Taken together, the replication and extension reinforce the importance of committee popularity in shaping representativeness, while also suggesting that its effect may be conditioned by institutional context. Future research could build on this by further investigating interaction effects and employing larger datasets or additional cases to more rigorously test these relationships.